HACKBOATS®

Center of Excellence in Robotics & Intelligent Systems

Industry Ready Curriculum — Engineered for the Future of Automation

3-YEAR PROGRAM · 360 HOURS · 6 SEMESTERS

About HackBoats

HackBoats stands at the forefront of innovation in engineering education, bringing over **12+ years** of specialized expertise in product development, transforming how students learn and apply cutting-edge technologies. Our comprehensive approach encompasses **Robotics, Artificial Intelligence, Drone Technology, Internet of Things (IoT), Product Development, and advanced 3D Printing solutions.**

12+

Years of Experience

Proven track record in engineering education transformation

200+

Workshops Conducted

Across multiple engineering colleges in India

27+

College Partners

Building innovation ecosystems nationwide

150K+

Students Trained

Transformed into industry-ready engineers

With a proven track record spanning the length and breadth of India, we've helped thousands of students transition from theoretical knowledge to practical, industry-ready skills. Our experienced team of engineers, innovators, and educators works tirelessly to bridge the gap between academic learning and real-world application, making every engineering student truly competitive in today's technology-driven job market.

> **Our Mission:** "To make every engineering student industry-ready by turning colleges into innovation hubs."

About HackBoats

HackBoats stands at the forefront of innovation in engineering education, bringing **12+ years of expertise in product development** and transforming how students learn and apply emerging technologies. Our work spans **Robotics, Artificial Intelligence, Drone Technology, Internet of Things (IoT), Product Development, and advanced 3D Printing**, delivered through **hands-on, industry-aligned programs** in collaboration with engineering colleges across India.

With a strong track record nationwide, HackBoats has helped thousands of students move from theoretical knowledge to practical, industry-ready skills. Our team of engineers and educators bridges the gap between academics and real-world applications, equipping students with technical depth, practical experience, and innovative problem-solving abilities through programs that blend core concepts, hands-on labs, and real-world project development.

HackBoats at a Glance

Industry-Aligned Programs

Curriculum co-designed with automation and robotics industry leaders

Hands-On First Philosophy

60% of all learning hours dedicated to lab and practical work

College Partnerships

Deployed as Center of Excellence programs in engineering institutions

Career Outcomes Focus

Graduates placed across robotics, automation, and AI sectors

Vision of the Center of Excellence

The Center of Excellence in Robotics & Intelligent Systems is envisioned as a world-class hub where students transform from theoretical learners into industry-ready robotics engineers. It serves as the nerve center for innovation, experimentation, and applied learning within the engineering institution.

Mission

To cultivate a generation of robotics engineers equipped with practical skills, computational thinking, and the adaptability to thrive in rapidly evolving automation industries.

Vision

To be the premier academic center for robotics education, producing graduates recognized for technical excellence, innovation, and industry readiness.

Industry Bridge

To forge deep, sustained partnerships with robotics and automation companies, ensuring the curriculum remains current, relevant, and professionally validated.

Academic Excellence

To meet and exceed the benchmarks set by NAAC, NIRF, and national accreditation bodies through documented outcomes, structured assessments, and continuous improvement cycles.

Industry Need for Robotics Engineers

The global robotics and automation market is projected to surpass **$214 billion by 2030**, driven by manufacturing modernization, autonomous systems, AI integration, and smart infrastructure. Yet a critical skills gap persists — industry leaders consistently report a severe shortage of engineers capable of designing, programming, and maintaining intelligent robotic systems.

3.5M

Unfilled Jobs

Robotics & automation roles projected globally by 2025

$214B

Market Size

Global robotics industry valuation by 2030

40%

Skills Gap

Of manufacturers report difficulty hiring qualified robotics engineers

High-Demand Career Sectors

🏭 Industrial Automation

Smart factories, CNC integration, PLC systems

🚗 Autonomous Vehicles

Self-driving systems, sensor fusion, path planning

🤖 Collaborative Robotics

Human-robot interaction, cobots in manufacturing

🛸 Drone Technology

UAVs for surveillance, delivery, and agriculture

🧠 AI & Computer Vision

Machine perception, deep learning, object detection

Program Structure Overview — 3 Years / 6 Semesters

The curriculum is designed as a progressive, scaffolded journey — each semester builds upon the last, deepening technical competency while expanding applied skills. Students move from foundational electronics and embedded systems through to AI-powered robotics and ROS-based advanced systems.

1. Semester 1
   Foundations of Robotics & Embedded Systems
2. Semester 2
   Embedded Systems & IoT Robotics
3. Semester 3
   Autonomous Robotics & Navigation
4. Semester 4
   AI Robotics & Computer Vision
5. Semester 5
   Industrial Robotics & Automation
6. Semester 6
   Advanced Robotics & ROS Systems

Semester	Hours	Year	Primary Focus	Key Platforms
Semester 1 — Foundations	60 hrs	Year 1	Electronics & Arduino	Arduino, Sensors
Semester 2 — Embedded & IoT	60 hrs	Year 1	ESP32, Wireless Control	ESP32, Mobile Robots
Semester 3 — Autonomous Nav.	60 hrs	Year 2	Sensor Fusion,Path Planning	Raspberry Pi, Line Follower
Semester 4 — AI & Vision	60 hrs	Year 2	OpenCV, Deep Learning	AI Kits, Cameras
Semester 5 — Industrial	60 hrs	Year 3	Robotic Arms, Cobots	Industrial Arms, PLC
Semester 6 — ROS & Advanced	60 hrs	Year 3	ROS, SLAM, Simulation	ROS, Quadruped, Tracked

Learning Model: Theory + Lab + Projects

Time Distribution Per Semester

Every 60-hour semester is structured to maximize practical skill acquisition while grounding students in the necessary theoretical frameworks.

30%

Theory

18 hrs — Concepts, principles, technical lectures

60%

Lab Work

36 hrs — Hands-on builds, experiments, prototyping

10%

Projects

6 hrs — Mini projects, evaluations, demonstrations

Pedagogical Philosophy

1 Concept Introduction

Each module opens with focused theoretical sessions covering core principles — electronics, programming logic, control theory, or AI concepts — delivered through interactive lectures and multimedia content.

2 Guided Lab Practice

Students immediately apply new concepts in structured lab sessions using real hardware — building circuits, programming microcontrollers, assembling robots, and running experiments in our purpose-built testing arenas.

3 Project-Based Evaluation

Each semester concludes with a mini-project and structured evaluation. Students demonstrate competency through working prototypes, technical reports, and live demonstrations — mirroring real industry deliverables.

Robotics Infrastructure & Lab Facilities

🏎️ Autonomous Driving Track

A scaled road network with lane markings, intersections, and traffic elements for testing autonomous vehicle algorithms and sensor-based navigation.

🧩 Robotics Maze Track

A reconfigurable maze environment used for testing maze-solving algorithms, flood-fill logic, and sensor-based wall-following navigation.

⚽ Robo Soccer Arena

A regulation-style arena for multi-robot coordination challenges, real-time decision making, and competitive team robotics events.

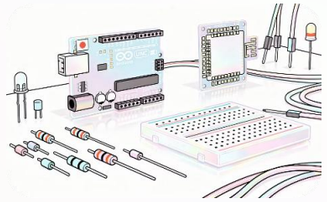
🚧 Obstacle Navigation Track

A dynamic obstacle course for testing collision avoidance, terrain adaptation, and sensor-motor integration on mobile and tracked robots.

The lab is equipped with individual workstations, component libraries, 3D printing facilities, soldering stations, and a dedicated electronics prototyping bench — providing students with a complete engineering development environment from concept to tested prototype.

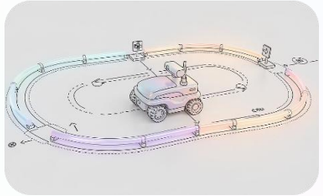
Hardware Platforms Used

The program leverages a carefully selected ecosystem of hardware platforms spanning from entry-level microcontrollers to advanced collaborative robotic systems — ensuring students develop competency across the full spectrum of modern robotics hardware.



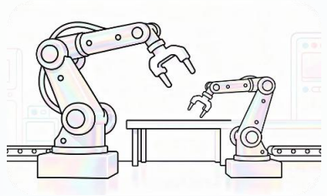
Microcontrollers

- Arduino Uno / Mega / Nano
- ESP32 / ESP8266 (Wi-Fi + BT)
- Raspberry Pi 4 / Zero



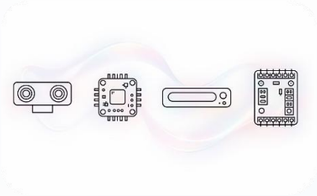
Robotics Platforms

- Mobile wheeled robots
- Line follower robots
- Maze-solving robots
- Soccer robots
- Autonomous vehicles



Advanced Systems

- Industrial robotic arms
- Collaborative robots (cobots)
- Autonomous tracked robots
- Quadruped robots
- AI robotics kits



Sensors & Components

- Ultrasonic, IMU, Cameras
- LiDAR (conceptual)
- Motor drivers, Encoders
- Servo & Stepper motors

Software Platforms Used

1

Arduino IDE

The primary development environment for Semesters 1–2. Students write, compile, and upload C/C++ programs to Arduino and compatible microcontrollers, building foundational embedded programming skills.

2

Python

Introduced in Semester 2 and used throughout the program. Python is the backbone of AI, computer vision, and ROS programming — critical for modern robotics development across both academic and industrial contexts.

3

ROS (Robot Operating System)

The industry-standard middleware for robotics. Covered in depth during Semesters 5 and 6, ROS enables students to build modular robotic systems using nodes, topics, services, and the comprehensive ROS toolchain.

1

OpenCV

The leading open-source computer vision library, used in Semester 4 to implement image processing, object detection, color tracking, and real-time visual feedback systems integrated into robotic platforms.

2

Simulation Tools

Gazebo and Webots simulation environments allow students to design, test, and iterate on robotic systems virtually before deploying to physical hardware — reducing risk and accelerating the development cycle.

3

Supporting Tools

Git for version control, Jupyter Notebooks for AI/ML experimentation, FreeCAD/Fusion 360 for mechanical design, and MATLAB (conceptual) for control systems analysis round out the software toolkit.

YEAR 1 · SEMESTER 1

Semester 1: Foundations of Robotics & Embedded Systems

Learning Objectives: Students will understand fundamental electronics principles, gain proficiency in Arduino programming, and build their first functional robots using sensors and actuators.

Modules & Topics Covered

Module	Key Topics
M1: Electronics Basics	Voltage, current, resistance, Ohm's Law, breadboarding, LEDs, resistors, capacitors
M2: Arduino Programming	Arduino IDE setup, digital/analog I/O, PWM, Serial communication, C++ fundamentals
M3: Sensors & Actuators	Ultrasonic sensors, IR sensors, servo motors, DC motors, motor driver L298N
M4: Basic Robotics	Robot anatomy, locomotion types, chassis design, wheel geometry, power systems
M5: Intro to Robotics Logic	Decision trees, conditional logic, basic state machines, sensor-driven behavior

Lab Activities & Mini Projects

LED & Sensor Circuit Builds

Breadboard circuits with LEDs, buttons, and ultrasonic distance measurement displays

Motor Control Lab

Program DC motors with L298N, implement speed and direction control via PWM

Obstacle Avoidance Bot

Mini Project: Build and program a robot that detects and avoids obstacles autonomously using ultrasonic sensors

Skills Developed: Circuit design, embedded C programming, sensor integration, basic robot mechanics, systematic debugging

YEAR 1 · SEMESTER 2

Semester 2: Embedded Systems & IoT Robotics

Learning Objectives: Students will master wireless communication protocols, develop IoT-connected robotic systems using ESP32, and build their first remote-controlled and internet-enabled robots.

Modules & Topics Covered

Module	Key Topics
M1: ESP32 Deep Dive	ESP32 architecture, GPIO programming, ADC/DAC, Wi-Fi & Bluetooth configuration
M2: Wireless Robotics	Bluetooth control, Wi-Fi web server control, RF modules, smartphone integration
M3: IoT Fundamentals	MQTT protocol, cloud dashboards, sensor data logging, Blynk / ThingSpeak
M4: Python for Robotics	Python syntax, GPIO with RPi.GPIO, serial communication, data parsing
M5: Line Following Logic	PID control theory introduction, IR sensor arrays, line detection, calibration
M6: Mobile Robot Systems	Differential drive kinematics, encoder feedback, speed calibration, turning logic

Lab Activities & Mini Projects

Wi-Fi Controlled Robot

Build a robot controlled via a web browser on the local network using ESP32 WebServer

IoT Sensor Dashboard

Stream real-time sensor data to a cloud dashboard using MQTT and ThingSpeak

Line Follower Robot

Mini Project: Implement a PID-tuned line follower capable of navigating standard competition tracks

Skills Developed: Wireless protocols, IoT integration, Python programming, PID tuning, mobile robot control, cloud connectivity

YEAR 2 · SEMESTER 3

Semester 3: Autonomous Robotics & Navigation

Learning Objectives: Students will implement sensor fusion techniques, develop path-planning algorithms, and build fully autonomous robots capable of navigating complex environments without human intervention.

Modules & Topics Covered

Module	Key Topics
M1: Sensor Fusion	Combining ultrasonic + IR + IMU data, Kalman filter basics, noise reduction
M2: Navigation Algorithms	Wall following, flood fill algorithm, Dijkstra's path planning, A* search
M3: Raspberry Pi Integration	Linux for robotics, GPIO via Python, Camera module, SSH remote access
M4: Encoder & Odometry	Wheel encoders, dead reckoning, odometry calculations, pose estimation
M5: Maze Solving	Maze mapping, memory-based navigation, solving strategies, tournament rules
M6: Autonomous Vehicles Intro	Lane detection basics, marker-based navigation, stop logic, traffic rules

Lab Activities & Mini Projects

Multi-Sensor Fusion Lab

Combine IMU and ultrasonic data to improve navigation stability and obstacle detection accuracy

Odometry Calibration

Implement encoder-based dead reckoning; validate pose estimates across measured distances

Maze Solving Robot

Mini Project: Deploy a flood-fill algorithm on the robotics maze track for fastest-path navigation

Skills Developed: Sensor fusion, autonomous navigation, Linux/RPi, path planning algorithms, odometry, competition robotics

YEAR 2 · SEMESTER 4

Semester 4: AI Robotics & Computer Vision

Learning Objectives: Students will apply machine learning concepts to robotics, implement real-time computer vision using OpenCV, and build AI-powered robots capable of recognizing, classifying, and reacting to visual stimuli.

Modules & Topics Covered

Module	Key Topics
M1: Python for AI	NumPy, OpenCV installation, image arrays, color spaces, thresholding
M2: Computer Vision	Object detection, color tracking, contour analysis, edge detection (Canny), HSV masking
M3: ML for Robotics	Classification basics, TensorFlow Lite, gesture recognition, model deployment on RPi
M4: Vision-Guided Robots	Camera-robot integration, visual servoing, target following, lane detection with OpenCV
M5: AI Robotics Kits	Pre-built AI kits, face detection, QR code navigation, voice command integration
M6: Real-Time Processing	Frame rate optimization, pipeline architecture, multithreading in Python for robotics

Lab Activities & Mini Projects

Color Tracking Robot

Program a robot to follow a colored object in real time using OpenCV HSV masking and PID control

Face & Object Detection

Deploy a pre-trained neural network on Raspberry Pi to detect and classify objects in the robot's camera feed

Vision-Guided Autonomous Bot

Mini Project: Build a robot that navigates autonomously using camera-based lane detection on the driving track

Skills Developed: OpenCV, computer vision, TensorFlow Lite, Python AI pipelines, real-time image processing, AI-robot integration

YEAR 3 · SEMESTER 5

Semester 5: Industrial Robotics & Automation

Learning Objectives: Students will operate and program industrial robotic arms and collaborative robots, understand automation system architecture, and apply robotics in simulated manufacturing environments.

Modules & Topics Covered

Module	Key Topics
M1: Industrial Robotics	DOF analysis, joint types, kinematic chains, workspace analysis, robot classification
M2: Forward & Inverse Kinematics	Transformation matrices, Denavit-Hartenberg parameters, IK solvers, trajectory planning
M3: Robotic Arm Programming	Teach pendant operation, waypoint programming, pick-and-place sequences, gripper control
M4: Collaborative Robots (Cobots)	Safety standards (ISO 10218), force/torque sensing, human-robot interaction design
M5: Automation Systems	PLC basics, ladder logic, sensors in automation, conveyor integration, SCADA overview
M6: Manufacturing Simulation	Digital twin concepts, simulation-to-real transfer, cycle time optimization

Lab Activities & Mini Projects

Robotic Arm Pick-and-Place

Program the industrial arm to perform precise pick-and-place operations on a simulated assembly line

Cobot Safety Integration

Configure a collaborative robot with safety zones, force limits, and human-proximity response behaviors

Automated Sorting Station

Mini Project: Design and implement a camera-guided sorting station using a robotic arm and conveyor

Skills Developed: Industrial robot programming, kinematics, cobot operation, PLC basics, automation system design, manufacturing simulation

YEAR 3 · SEMESTER 6

Semester 6: Advanced Robotics & ROS Systems

Learning Objectives: Students will master the Robot Operating System (ROS) architecture, implement SLAM and advanced navigation, and develop integrated robotic systems using quadruped and tracked autonomous robots.

Modules & Topics Covered

Module	Key Topics
M1: ROS Architecture	Nodes, topics, services, actions, roslaunch, catkin build system, ROS2 overview
M2: SLAM & Mapping	SLAM theory (gmapping, cartographer), occupancy grids, map building with LiDAR/camera
M3: Advanced Navigation	Navigation stack (move_base), costmaps, AMCL localization, dynamic obstacle avoidance
M4: Quadruped Robotics	Quadruped gait analysis, body IK, terrain adaptation, balance control principles
M5: Autonomous Tracked Robots	Track-based locomotion, all-terrain navigation, remote-supervised autonomy
M6: Capstone Integration	Full system integration, ROS-based capstone development, documentation, presentation

Lab Activities & Capstone

ROS Navigation Stack

Deploy a mobile robot with full autonomous navigation using ROS move_base, costmaps, and AMCL localization

SLAM Mapping Lab

Build a real-time map of the robotics arena using the quadruped or tracked robot and gmapping

Capstone System Build

Capstone Project: Design, build, and demonstrate a fully integrated autonomous robotic system solving a defined engineering challenge using ROS

Skills Developed: ROS development, SLAM, advanced navigation, quadruped programming, full system integration, technical documentation

Competitions

Throughout the 3-year program, students participate in escalating levels of competitive and collaborative challenges designed to simulate real engineering environments and build professional confidence.

🏁 Line Follower Challenge

Semester 2 competition using the official line follower track. Teams compete on speed, accuracy, and PID tuning quality. Evaluated on lap time and path adherence.

🧩 Maze Solving Tournament

Semester 3 challenge on the robotics maze track. Students compete for fastest full-maze solve using implemented flood-fill or custom algorithms with full documentation review.

⚽ Robo Soccer League

Semester 4 multi-robot coordination event in the Robo Soccer Arena. Teams of 2–4 robots compete in structured match play, testing vision, coordination, and decision logic.

🚗 Autonomous Driving Challenge

Semester 5–6 capstone-level competition on the autonomous driving track. Robots must perform lane keeping, obstacle avoidance, traffic sign recognition, and parking.

Many more.....

Capstone Project Framework

Evaluation Criteria

30%

Technical Build

Functionality, reliability, and engineering quality of the robot system

25%

Innovation

Originality of approach and problem-solving methodology

25%

Documentation

Technical report, circuit diagrams, code, and design rationale

20%

The Semester 6 capstone project represents the culmination of the entire 3-year curriculum. Students form teams of 3–4, select a real-world engineering problem, and design, build, and deploy a fully integrated robotic solution — from concept through validated testing and professional presentation.

Capstone Project Tracks

Track A: Industrial Automation

Design an automated robotic workstation for a defined manufacturing task using a robotic arm, vision system, and PLC logic

Track B: Autonomous Navigation

Build a fully autonomous mobile robot capable of mapping an unknown environment, localizing, and completing a delivery task using ROS and SLAM

Track C: AI-Integrated Robot

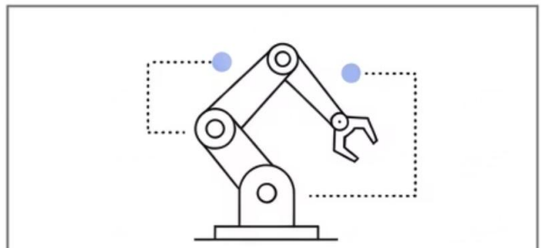
Develop a robot that uses computer vision and machine learning to detect, classify, and respond to real-world visual conditions in a dynamic environment

Track D: Open Innovation

Students propose an original robotics solution addressing a real community or industry challenge — subject to faculty and industry panel approval

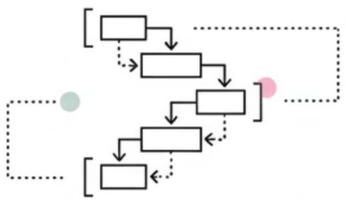
Skills Developed Through the Program

Upon completing all 6 semesters, graduates will possess a comprehensive, layered skill set spanning hardware engineering, software development, intelligent systems, and professional practice — fully aligned with industry hiring benchmarks.



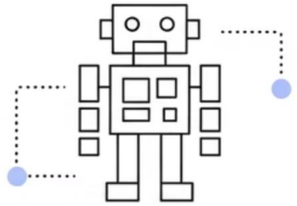
HARDWARE SKILLS

Circuits, Microcontrollers, Assembly



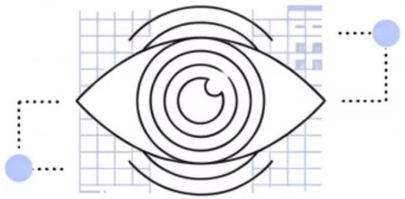
SOFTWARE SKILLS

Python, ROS, OpenCV, Git



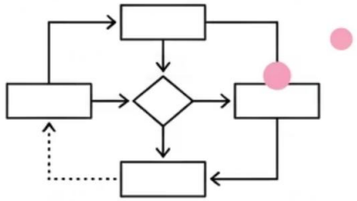
ROBOTICS ENGINEERING

Kinematics, PID, SLAM, Navigation



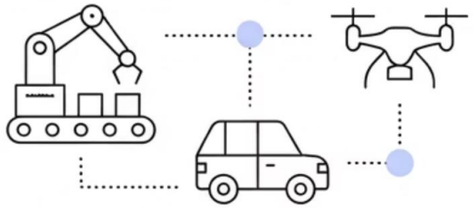
AI & VISION

Vision, Detection, ML, TensorFlow



PROFESSIONAL SKILLS

Documentation, Teamwork, Presentation, Problem-Solving



INDUSTRY APPLICATIONS

Automation, Cobots, Drones, Integration

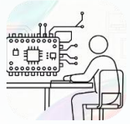
6	360	20+	10+
Skill Domains	Training Hours	Hands-On Projects	Platforms Mastered
Hardware, software, AI, robotics, professional practice, industry applications	Total hours of structured learning across 3 years and 6 semesters	Mini projects, competition challenges, and capstone deliverables completed	Hardware and software platforms across the full robotics development stack

Career Pathways & Industry Opportunities

Graduates of the Center of Excellence in Robotics & Intelligent Systems are prepared for a diverse range of high-demand, high-compensation careers across the global automation and technology landscape.

Robotics & Industrial Automation Engineer

Design, deploy, and maintain robotic systems in manufacturing plants, smart factories, and automated logistics facilities. Work with PLCs, cobots, and robotic arms across sectors including automotive, electronics, and pharmaceuticals.



Embedded Systems & IoT Developer

Develop firmware and connected device systems for robotics, consumer electronics, smart infrastructure, and industrial IoT. Strong demand in automotive, aerospace, telecommunications, and smart energy sectors.

Autonomous Systems & Drone Engineer

Engineer navigation, sensing, and control systems for autonomous vehicles, UAVs, and unmanned platforms in defense, agriculture, logistics, and urban mobility sectors globally.

AI Robotics & Computer Vision Engineer

Apply machine learning and visual perception to robotic systems — building intelligent machines that see, recognize, and respond. High demand in quality inspection, medical robotics, retail automation, and smart surveillance.

Conclusion & Impact of the Center of Excellence

The Center of Excellence in Robotics & Intelligent Systems, powered by HackBoats, represents a transformative investment in engineering education. By combining a rigorous 360-hour, 6-semester curriculum with world-class hardware infrastructure, industry-validated content, and a hands-on-first pedagogy, this program bridges the critical gap between academic preparation and professional readiness.

Students who complete this program will graduate not only with technical credentials, but with demonstrated, portfolio-backed competency in robotics engineering, embedded systems, AI, and autonomous systems — making them among the most industry-ready graduates in the country.

📋 For NAAC / NIRF

Documented outcomes, structured assessments, and industry linkages supporting accreditation benchmarks

🤝 For Industry Partners

A pipeline of trained, hands-on robotics engineers ready for direct employment and collaborative R&D

🎓 For Students

A career-defining program that builds real skills, real confidence, and real professional opportunities

Program at a Glance

01

3 Years · 6 Semesters · 360 Hours

Comprehensive, progressive curriculum from fundamentals to advanced systems

02

60% Hands-On Lab Learning

Students spend the majority of their time building, testing, and iterating with real hardware

ćċ

7 Career Pathways Covered

Robotics, automation, embedded, AI, autonomous, drones, and mechatronics

ćç

Industry-Ready Graduates

Validated by competitive challenges, capstone projects, and professional documentation standards

> *"Engineering education must evolve alongside industry. This Center of Excellence ensures our students lead that evolution — not follow it."*

Contact Information

Get in Touch

We're eager to discuss how the HackBoats Centre of Excellence can be customized for your institution's specific needs and goals. Our team is available to answer questions, provide additional information, and schedule campus visits at your convenience.

Whether you're a institution looking to enhance your college's reputation, seeking to improve department outcomes, or a faculty member passionate about student success, we welcome the opportunity to explore this partnership with you.

Website

https://hackboats.com/

Contact number

888-60-999-57

Likhith –Robotics Engineer & Senior Business Development Manager.

Location

Hyderabad, India

Next Steps

Call Us

Schedule a consultation call to discuss your specific requirements and timeline

Campus Visit

Arrange for our team to visit your college and present detailed proposal to stakeholders

Partnership Begins

Sign MoU and launch your Centre of Excellence, transforming student futures